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Industrializing test assets in the GenAI era

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THE OPENING

Why industrialization became my fascination

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- Led large-scale tech & test modernization programs
- Specialized in test asset governance and enhancement
- Focused on sustainable, scalable quality engineering

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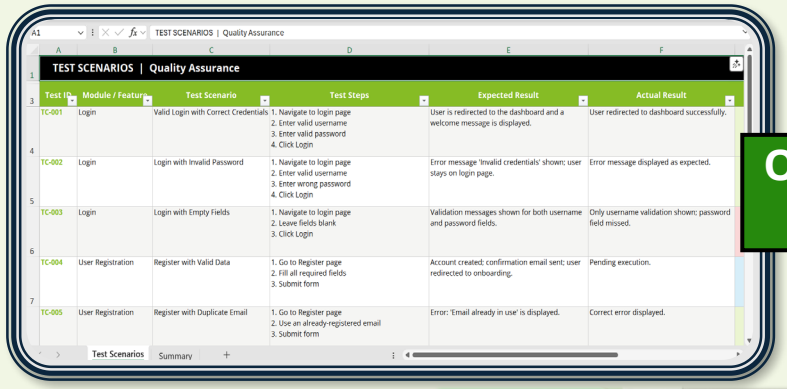
THE OPENING

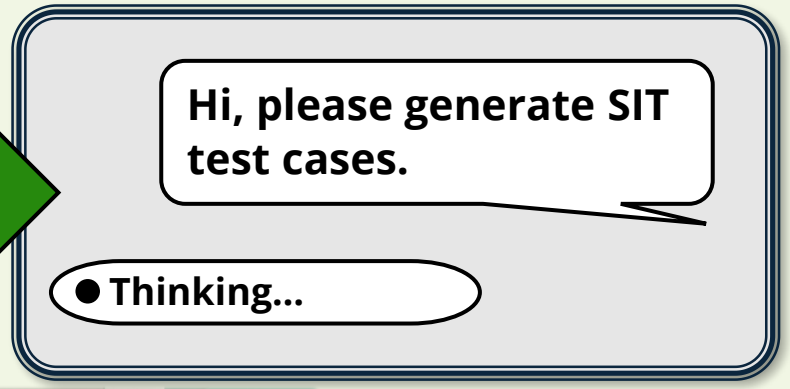
The tipping point



TEST SCENARIOS | Quality Assurance

Test ID	Module / Feature	Test Scenario	Test Steps	Expected Result	Actual Result
TC-001	Login	Valid Login with Correct Credentials	1. Navigate to login page 2. Enter valid username 3. Enter valid password 4. Click Login	User is redirected to the dashboard and a welcome message is displayed.	User redirected to dashboard successfully.
TC-002	Login	Login with Invalid Password	1. Navigate to login page 2. Enter valid username 3. Enter wrong password 4. Click Login	Error message 'Invalid credentials' shown; user stays on login page.	Error message displayed as expected.
TC-003	Login	Login with Empty Fields	1. Navigate to login page 2. Leave fields blank 3. Click Login	Validation messages shown for both username and password fields.	Only username validation shown; password field missed.
TC-004	User Registration	Register with Valid Data	1. Go to Register page 2. Fill all required fields 3. Submit form	Account created; confirmation email sent; user redirected to onboarding.	Pending execution.
TC-005	User Registration	Register with Duplicate Email	1. Go to Register page 2. Use an already-registered email 3. Submit form	Error: 'Email already in use' is displayed.	Correct error displayed.

Output in a Day



5-10 TEST CASES

1000++ TEST CASES

The line gets crossed when test growth outpaces a team's ability to keep tests trustworthy

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THE OPENING

The avalanche of test cases

Increased velocity

Businesses are generating more tests, faster than ever before

The consequence

This speed, without structure, can lead to a slower and less trusted regression, potentially resulting in releases that either delay or ship with blind spots

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The Problem

When acceleration creates instability

Test volume is scaling faster than test trust and that erodes release confidence.

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THE PROBLEM

The legacy debt problem

What happens when test cases outlive the context that created them?

Structurally unmappable

Inherited suites carry no traceable link to requirements or risk areas.

No exit strategy

Tests are never retired. Obsolete scenarios accumulate

Value decays silently

Maintenance cost persists long after protective value expires

Many companies continue paying to maintain assets that no longer reduce risk, and it often goes unnoticed until the gap is visible

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THE PROBLEM

How organizations can make it worse

What can happen when teams actively compound the debt they inherit?

Coupled & fragile

Renames can cascade into broken tests. Scripts are cloned rather fixed.

Locked & obsolete

Tests tied to one environment and written against requirements that change mid-way.

Entangled & accelerating

Shared test data and order-dependent execution hide failures. Speed hide coverage gaps.

Automation output rises, but defect leakage and release delays may persist. Speed without structure is often just faster failure.

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THE PROBLEM

The automation paradox

Leaders

The impact gap

- Escalating costs without reduced risk
- More automation, longer delays
- Paying for activity, not impact

Practitioners

Maintenance burnout

- Brittle tests & constant fixes
- Copy-paste scripts create hidden debt
- The reward for more work is... More work

Buyers

Diminishing returns

- Paying for test volume, not value
- Mistaking tooling for strategy
- Weaker performance contribution

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The Model

Industrialized test assets at scale

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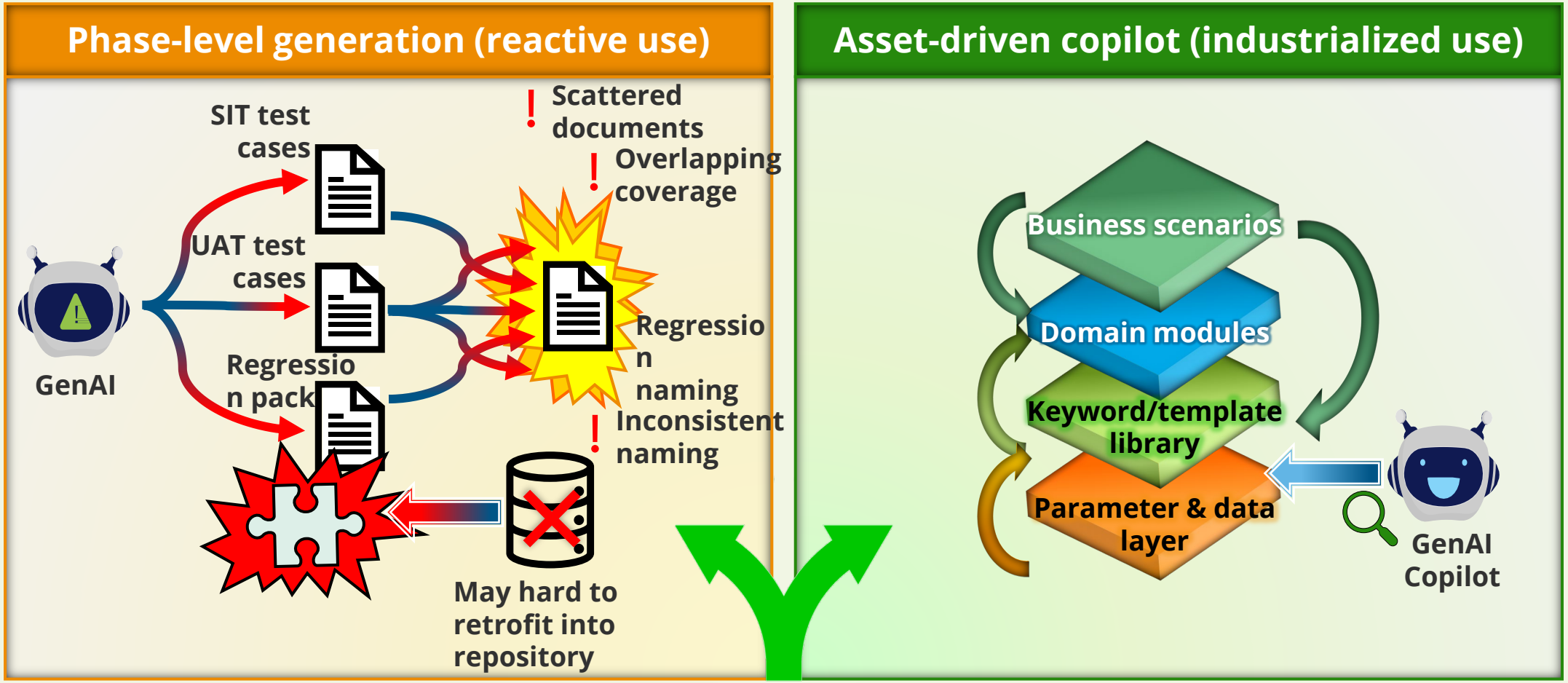
THE MODEL

Where GenAI operates matters



Fast creation. Slow stabilization

Structure acceleration. Sustainable scaling

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THE MODEL

GenAI acts as co-pilot, not a factory

A controlled model: from inputs to promotion

1. Inputs matter (The anchor)

- Templates
- Glossary
- Conventions
- Schema

Context anchors output

No blank prompts

2. Constrained generation (The candidate)

NOT for entire packs, free-form scenarios Or bypassing architecture

3. Validation & promotion (The discipline)

- ☑ Conform
- ☑ No duplicate
- ☑ Be reviewed

Disciplined promotion

Nothing enters automatically

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Detecting misplaced acceleration

Rising redundancy

Similar logic across assets, declining template reuse ratio

Increasing flakiness

Hard-coded test data, environment-sensitive assertions

Unclear test intent

Long narrative-style steps, mixed scenario & template logic, weak traceability to reusable components

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THE MODEL

Three pillars of test industrialization

Standardization

- ✓ Structure
- ✓ Templates
- ✓ Intent clarity

Optimization& reuse

- ✓ Modularity
- ✓ Abstraction
- ✓ Parameterization

Governance

- ✓ Ownership
- ✓ Decision signals
- ✓ Lifecycle control

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Pillar 1: Standardization

Designing for clarity and consistency

Design standards

- Structure by capability, not by phase
- Speak the system's language
- Remove artificial anchors

What this achieves

- Predictability
- Readability
- Cognitive efficiency
- AI alignment

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Pillar 2: Optimization & reuse

Focus on the structural level, not at the execution level

Reuse mechanics

- Keyword-driven/modular building block
- Parameterization
- Create, read, update, delete as a coverage accelerator

What this achieves

- Controlled test growth
- Single-point maintenance
- Coverage without sprawl
- Structural scalability

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Pillar 3: Governance

Keep tests healthy at scale

Control points

- Defined test operating model
- Roles & permissions as quality control
- Centralized, shareable reporting

What this achieves

- Clear asset accountability
- Managed asset lifecycle
- Stable regression over time
- Early detection of test sprawl

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The Tension

When industrialization *goes too far*

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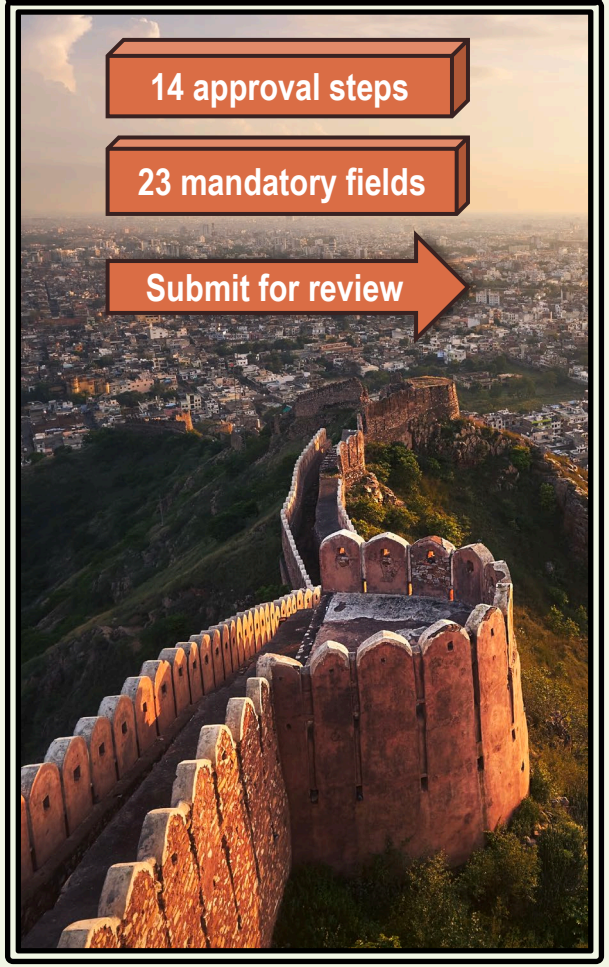
THE TENSION

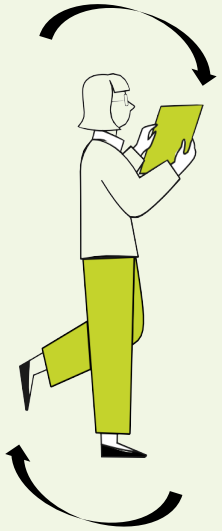
When good intentions go rogue

The abstraction tower

The fixture jungle

The governance fortress





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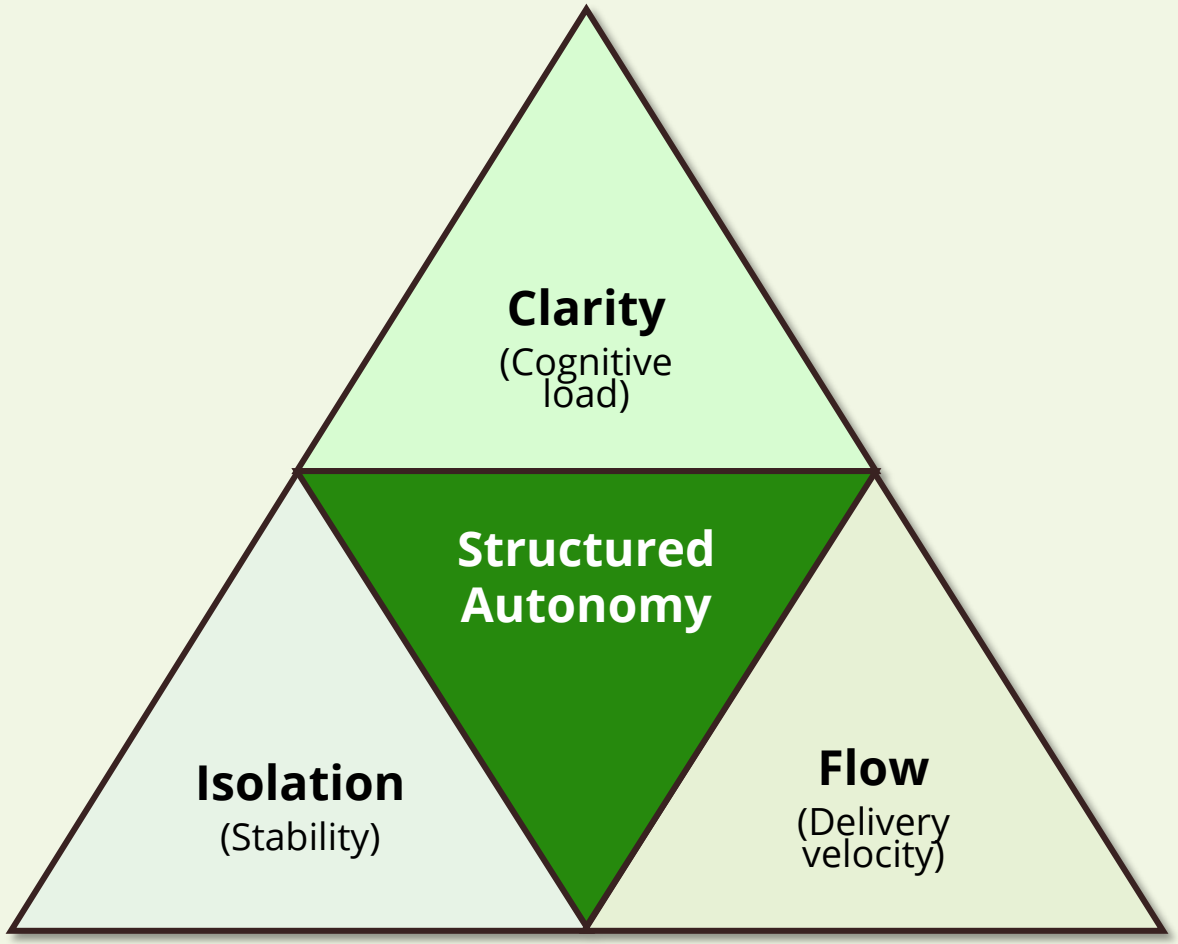
THE TENSION

The balance rule



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The Measurement

Measurement as decision signals

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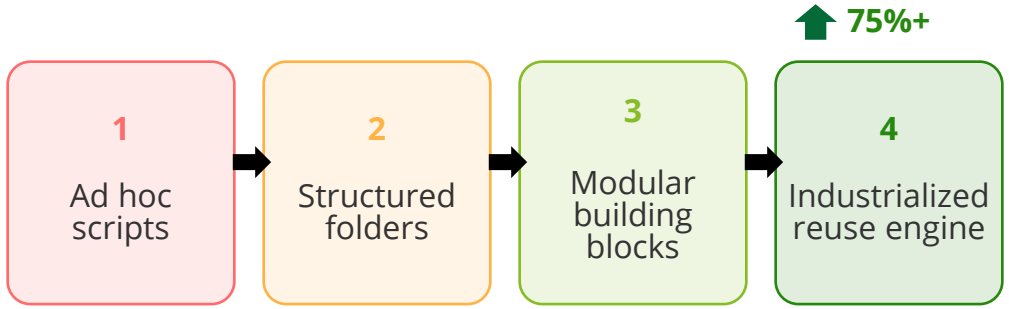
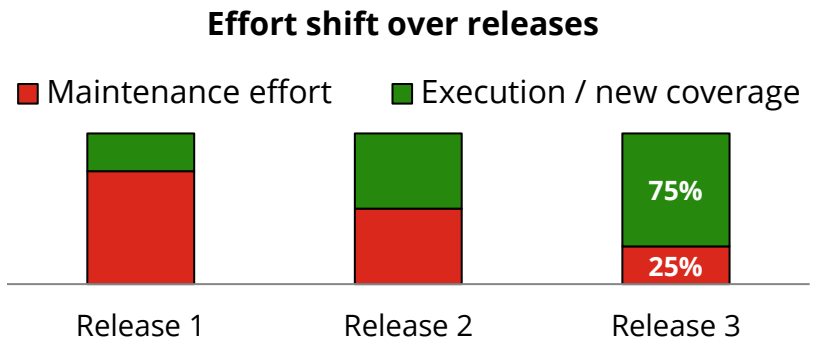
Metrics vs decision signals

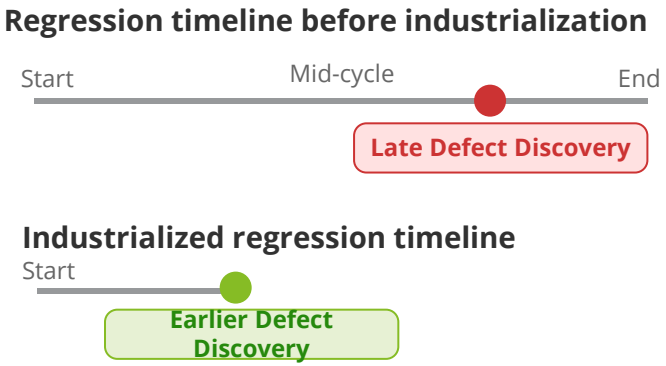
Reuse ratio are we scaling effectively?



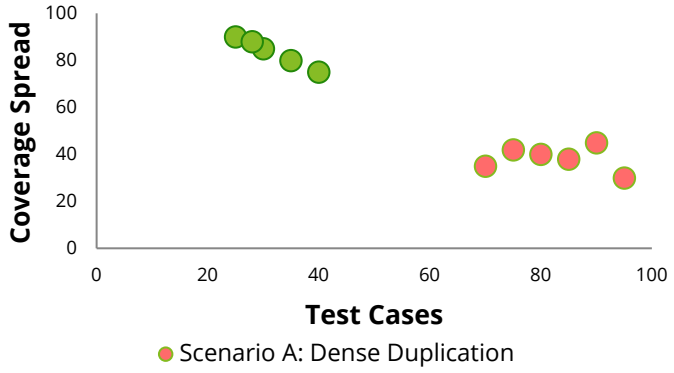
Maintenance load shifting from fixing to building?



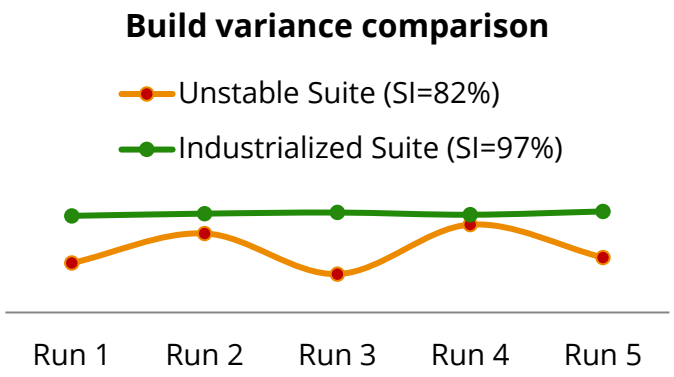
Regression half-life fast feedback?



Coverage density is volume misleading?



Stability index can we trust results?



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Minimum viable signal set

Five signals, five decisions

SIGNAL	STATUS (Illustrative example)	DECISION
Reuse ratio	▶ 42%	Increase modular refactor
Stability index	▲ 96%	Maintain
Maintenance load	▼ 35%	Enhance templates
Regression half-life	▶ Trending up	Rationalize
Coverage density	▲ Balanced	Sustain

KEY: STATUS INDICATORS

- ▲ (Blue) Stable/Healthy
- ▶ (Yellow) Trending/Warning
- ▼ (Red) Action required

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From Insight to Action

What you can *do next*

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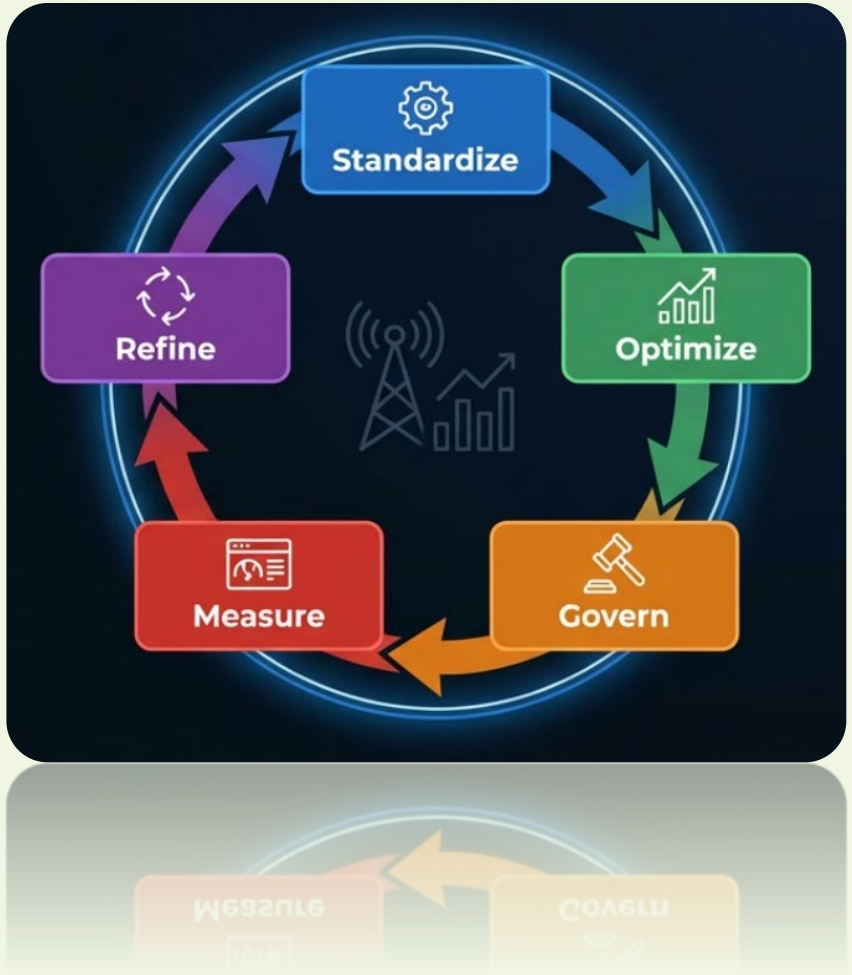
FROM INSIGHT TO ACTION

Three final takeaways

GenAI should expand coverage, not duplicate it

Reuse is a design decision, not an accident

Stability is a governance outcome, not a tooling feature



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FROM INSIGHT TO ACTION

Start industrializing

Your first sprint
A timeboxed kickoff

1
Run a timeboxed test asset health check

- Identify a sample of overlapping test cases
- Count how many shared identical steps
- Identify your most brittle tests
- Highlight duplicate AI-generated scripts

2
Identify one reusable building block

- Extract one common workflow
- Convert it into a modular template
- Parameterize test data instead of copying scripts

3
Establish one governance guardrail

- Define a naming convention
- Assign ownership to one test library
- Track one stability metric (e.g., regression failure volatility)

Industrialization started
Small actions compound. You've started the momentum.

Scale steps to your suite size and team maturity

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GenAI can accelerate test creation.

Industrialization is what can make that acceleration sustainable.

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Q&A

What are your thoughts?
Let’s discuss

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If you're serious about industrializing test assets… Please visit our booth or scan the QR code for my business card.

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